Kishan Munjpara

0450960469 | kishanmunjpara2710@gmail.com | KishanMunjpara | KishanMunjpara | Portfolio

Education

Master of Information Technology (AI) — **February 2026 - Present**
Macquarie University

B.Tech in Information and Communication Technology — **November 2020 - May 2024**
PDEU, Gandhinagar-Gujarat — CGPA: 9.67/10

***Coursework**: ML, Deep Learning, AI, Gen-AI NLP, DBMS, Computer Communication Networks, Quantum Computing, Computer Vision, Theory Of Computation, Data Science, SQL for Beginners, etc.*

Technical Skills

Programming: Python, FastAPI, SQL, Cypher
AI/ML & GenAI: PyTorch, TensorFlow, scikit-learn; Transformers (Hugging Face); LLM APIs and LangChain; retrieval / embedding workflows for text; YOLO / CNN-based workflows for vision
Data & Storage: SQL, Neo4j, MongoDB, Firestore; Azure Blob Storage, Amazon S3
Cloud & DevOps: Microsoft Azure, Docker, Git, GitHub Actions, Streamlit, Linux
Soft Skills: Time Management, Teamwork, Problem-solving, Research-oriented Documentation.

Professional Experience

Smart Valyou — **August 2024 - February 2026**
Data and AI Engineer (Promoted from Data Eng & AI Intern) — *Singapore-Remote*

- Contributed to an AI-driven product focused on pricing and decision support: researched approaches, implemented algorithmic components in Python, and iterated designs using large, real-world datasets.
- Engineered data workflows and graph-backed data access using Neo4j: modelled entities and relationships, implemented graph queries including ETL/load steps to support product features and analytics, and integrated graph data with Python services and Azure-hosted APIs.
- Built and refined data pipelines for ingest, cleaning, and feature preparation, improving reliability and scalability of downstream AI components and product-facing workflows.
- Applied NLP/LLM-based retrieval and Q&A over policy and business documents, with evaluation against baseline rules or models.

Learning: Neo4j; Python data/ML engineering; Azure API integration; product-oriented AI and pipeline design

Phenomenal AI — **July 2024 - February 2026**
AI Software Engineer (Promoted from Intern & Junior Engineer) — *Mumbai-Remote*

- Delivered features across a generative video platform (frontend, backend, and cloud), including user/session data in MongoDB, video artefacts on AWS S3, and service APIs orchestrating generation jobs.
- Improved an open-source video generation stack end-to-end (inference configuration, pipeline orchestration, and resource usage), raising output quality and cutting latency and compute cost for production use.
- Shipped ”Director”, an interactive assistant that helps users author and refine video scenes (prompt/scene design in-product), increasing engagement and the quality of inputs to generation.
- Used modern AI-assisted development workflows in Cursor to accelerate prototyping while keeping code review, testing, and release discipline for production features.

Learning: Text-to-Video generation, Azure Storage, Storage Optimization Techniques, MongoDB, S3.

iCEM - promoted by GMDC — **January 2024 - Jul 2024**
Intern — *Ahmedabad-Gujarat*

- Prototyped an integrated Computer Vision, RFID, and IoT solution to monitor mining yard traffic and vehicle movement, targeting ~50% faster turnaround and more reliable real-time tracking.
- Analysed historical sales and operational data, produced forecasting-oriented analytics and dashboard views, and translated findings into recommendations for marketing and planning.
- Worked with IT, sales, and project stakeholders to scope requirements, iterate prototypes, and align AI components with deployable mining operations workflows.
- Developed or supported an LLM-based assistant for financial report interpretation and Q&A, improving how users retrieved and validated information from long documents.

Learning: Enterprise Resource Planning (ERP), Data Analysis, IoT and RFID integration with AI

Amazon Summer School — **Sep 2023 – Oct 2023**
Mentee — *Remote*

- Learned Machine Learning (ML) topics including Supervised Learning, Deep Neural Networks, Dimensionality Reduction, Unsupervised Learning, Probabilistic Graphical Models, Sequential Learning, Causal Inference and Reinforcement Learning.

Learning: Foundations for applied deep learning and probabilistic modelling in real-world pipelines.

IIT RAM — **May 2023 - July 2023**
Summer Research Intern — *Ahmedabad-Gujarat*

- Developed a deep learning approach based on DeiT for estimating cattle age from dental images, including dataset preparation, training and validation protocols, and comparative experiments against baselines.
- Delivered a research prototype suitable for evaluation on mobile-captured images, supporting the team's filing of a related patent on livestock age estimation from dental imagery.

Learning: Image Processing, Computer Vision, Transformer Models

Projects

ChatBot: AI-powered Conversational Agent — **April 2024- June 2024**

- Implemented a real-time interactive chatbot featuring customizable system prompts and efficient memory management using the Meta-Llama-3-8B-Instruct model.
- Deployed the chatbot with Streamlit, ensuring seamless user interaction and effective chat history management.

Optimized Context Retrieval System for LLM-based Chatbots — **February 2024 - June 2024**

- Built a chatbot by integrating Dense Passage Retrieval with GPT and LLAMA, achieving precise context-based information retrieval.
- Enhanced the assistant's accuracy and understanding by utilizing advanced technologies, including the Meta LLAMA 2 model and Hugging Face Embeddings.

PalmIoT SecureLink | *Raspberry Pi4, Python...* — **August 2023 - November 2023**

- Developed a palm recognition system for security purposes, implementing feature extraction techniques such as HOG and LBP.
- Integrated the system into hardware using Raspberry Pi and an IP web camera setup by combining multiple convolutional neural networks and utilizing ensemble learning techniques.

Publication

Density Based Automatic Traffic Control System Using ML | IEEE — **March 2024**

- Engineered and deployed an advanced traffic control system using the YOLO machine learning algorithm. Attained a 40% reduction in traffic congestion during peak hours, leading to a 25% decrease in average commute time.
- Showcased the efficacy of this technology in optimizing traffic management, alleviating congestion, and enhancing road safety through real-time density analysis.

System and Method For Estimating Livestock Age Through Deep Learning| PATENT — **March 2024**

- The invention pertains to a system and method for accurately estimating the age of nearly 93% of livestock animals (cattle) by leveraging their dental images in a deep learning neural network (data transformer).
- This livestock dental image analysis system employs a self-learning model with 6 identical layers, operational on both user devices and servers, to analyze dental images captured via a mobile camera.

Awards and Certifications

- **NPTEL**: Constrained and Unconstrained Optimization (IIT-KGP).
- **Coursera**: Introduction to Statistics.(Stanford University)
- **ISRO**: Machine Learning to Deep Learning: A journey for remote sensing data classification.
- **AWS**: Demystifying AI/ML/DL.
- **Oracle Academy**: Database Programming with SQL.
- **IEEE Computational Intelligence Society**: Instructor at Quantum Computing Workshop, delivered a session on entanglement and teleportation for quantum computing.
- **NPTEL**: Probability Theory for Data Science.